

Complete Bounded Lozi Dynamics: A Global Symbolic Coding Theorem

HCS-C385 theorem and reproducibility package

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Abstract

For the area-preserving Lozi family $F_a(x, y) = (1 - a|x| - y, x)$ with $a > 4$, we reconstruct the entire set of bounded two-sided orbits by a uniform contraction on sequence space. Every binary sequence occurs once, and the whole bounded set stays a positive distance from the nonsmooth switching line. This resolves global completeness, not merely a finite periodic prefix.

Keywords: Lozi map; bounded orbit; symbolic coding; uniform hyperbolicity; primitive cycle; instability clock.

中文摘要

本文对参数大于四的保面积 Lozi 映射，利用双向序列空间上的一致压缩，重建全部有界轨道，并证明每个二进制符号序列恰好对应一个状态。有界不变集与不可微切换线具有严格正间隔，因而全局完备性不依赖有限周期外推。

关键词：Lozi映射；有界轨道；符号编码；一致双曲性；本原周期；不稳定性时钟。

1 A global question beyond finite word enumeration

Finite sign words can be solved exactly without showing that every infinite itinerary exists or that every bounded geometric orbit has been found. For a nonsmooth map, a coding must also resolve derivatives at its branch boundary. We close these questions in one explicit sufficient chamber of the globally defined map

$$F_a(x, y) = (1 - a|x| - y, x), \quad a > 4. \quad (1)$$

Its inverse is $(x, y) \mapsto (y, 1 - a|y| - x)$. Let K_a be the set of points whose full two-sided orbit is bounded. The source clock is one application of (1). We do not identify that clock with any later suspension.

Lozi symbolic dynamics and kneading methods have established owners [1, 2]. Our contribution is a directly proved, normalized chamber theorem with independent exact evidence; no priority for the horseshoe mechanism or cone method is claimed. Earlier project C116 certified a pruned finite prefix at $(a, b) = (2, 1/2)$ of $(1 - a|x| + by, x)$. It did not prove the present statement. The constant-diagonal affine horseshoe in C127 also does not provide the word-dependent derivative matrices below.

2 All bounded two-sided orbits

Put $\Sigma = \{-1, +1\}^{\mathbb{Z}}$, with the left shift σ , and define

$$R = \frac{1}{a-2}, \quad q = \frac{2}{a}, \quad \delta = \frac{a-4}{a(a-2)} > 0.$$

For $s \in \Sigma$ let

$$(\mathcal{J}_s x)_j = \frac{s_j}{a}(1 - x_{j-1} - x_{j+1}) \quad \text{on } \ell^\infty(\mathbb{Z}).$$

Theorem 1. *There is a homeomorphism $\pi_a : \Sigma \rightarrow K_a$ satisfying $F_a \pi_a = \pi_a \sigma$. Every coordinate in the coded orbit obeys $\delta \leq s_j x_j \leq R$. These are all bounded two-sided orbits in the plane.*

Proof. The operator $\mathcal{T}_s$ contracts by $q < 1$ and preserves the closed radius- R ball because $(1 + 2R)/a = R$. Its unique fixed sequence satisfies

$$s_j x_j = \frac{1 - x_{j-1} - x_{j+1}}{a} \geq \frac{1 - 2R}{a} = \delta.$$

Hence its signs are exactly s_j and it solves $x_{j+1} = 1 - a|x_j| - x_{j-1}$. Conversely, a bounded orbit with $M = \sup_j |x_j|$ satisfies $aM \leq 1 + 2M$, so $M \leq R$. The same sign bound applies, and contraction uniqueness recovers that orbit from its sign sequence.

Set $\pi_a(s) = (x_0, x_{-1})$. Starting the contraction at zero, the N th approximation to either coordinate depends only on a finite symbol window and has error at most Rq^N . This proves continuity in the product topology. A state determines the full orbit under the continuous inverse of F_a , so distinct sign sequences cannot map to the same state. Exhaustion gives surjectivity. A continuous bijection from compact Σ to its Hausdorff image is a homeomorphism; the recurrence proves shift equivariance. $\square$

The coding is also uniform in parameters: comparing the fixed-point equations for $a, b > 4$ yields

$$\|x_a(s) - x_b(s)\|_\infty \leq \frac{|a - b|}{(a - 2)(b - 2)}. \quad (2)$$

Indeed, after moving the $2/a$ difference term to the left, the remaining bound is $|a^{-1} - b^{-1}|(1 + 2/(b - 2))$. The chamber boundary is deliberately not called sharp: at $a = 4$ this argument has zero sign margin and supplies no conclusion.

3 Conclusion

The contraction theorem exhausts the global bounded set and resolves the switching-line issue in an explicit sufficient chamber.

References

- [1] M. Misiurewicz and S. Štimac, *Symbolic dynamics for Lozi maps*, arXiv:1601.01745 (2016). Primary preprint.
- [2] W.-M. Zheng and J.-x. Liu, Symbolic analysis of attractor geometry for the Lozi map, *Physical Review E* **50**, 3241 (1994). doi:10.1103/PhysRevE.50.3241.

Round zero: complete bounded coding.